AR Mine Digital Twin: Market, Customer and Business Case

Commercial strategy document · Draft v1.0 · October 2026 · Launch market: Ghana

Status of this document. This is a planning document built from the product owner's brief and chat messages only. No product specification, customer data, voice-of-customer research or web research was available. Facts are labelled by evidence tier (see Section 14). Every number marked as an assumption is a placeholder to be replaced with pilot data, not a finding. No market size or competitor facts are asserted.

---

1. Executive Summary

The opportunity. The Product is an augmented-reality digital twin of a mine. It builds a 3D model of the site, keeps it updated with live data, and lets several people view the same model at once, on AR headsets or on tablet, phone and web, on site or remote. A local edge node keeps sessions running where connectivity is poor and syncs to the cloud later.

Where to start. Three segments are prioritised, all subject to validation:

1. **Large open-pit gold producers in Ghana.** They have the most data, systems and budget, and the weekly multi-discipline planning review is a natural fit for shared AR.
2. **Underground operators.** They face the highest spatial complexity, and edge sync directly addresses their connectivity problem.
3. **Mine-service contractors and OEM/dealer service teams.** Remote expert assist has a measurable cash baseline (expert callouts) and a shorter sales cycle.

Most defensible value claim today. Several people, on headsets, tablets, phones or web, can view the same site model with live data in one shared session, including at sites with poor connectivity. The benefit is less travel and less explanation. This rests on the owner's stated features; no quantified saving is defensible until pilots produce measurements.

Key finding from the value model. Under illustrative base assumptions, no single use case covers an annual site licence of USD 36,000. Remote assist covers about 30% of it, planning review about 27% and HSE induction about 40%. To close the gap, the company must do one of three things: price the licence nearer USD 24,000, pitch several use cases per site, or lead with high-callout OEM and dealer customers. This is the first thing the pilots must test.

Recommended next actions

1. Share a specification or demo so evidence tiers can be upgraded.
2. Run 10–12 discovery interviews using the guide in Section 12.
3. Pilot the weekly planning review at one open-pit site, with remote assist as a parallel OEM pilot.
4. Measure baselines (review hours, callouts, induction time) before quoting any return on investment.
5. Define the integration scope, headset and device matrix, and edge-node specification.

Five biggest risks: feature status is unverified; data integration may decide adoption; headsets may not suit dust, heat, glare and underground conditions; safety-alert liability if latency or accuracy fails; and unproven willingness to pay against the licence price.

---

2. Product Overview

2.1 Value proposition

The Product gives mine teams a shared visual and collaboration layer on top of existing mine data. Everyone sees the same live 3D picture, whether standing in the pit or sitting in a head office.

2.2 What is known and what is assumed

Item	Known (stated by owner)	Planning assumption
AR digital twin	Stated	Model built from client survey, CAD and GIS files, refreshed by drone or LiDAR

Item	Known (stated by owner)	Planning assumption
Alignment accuracy	Unknown	Pilot target of 0.5 m or better for overview tasks; tighter only after testing
Real-time streaming	Stated	1–2 data sources per pilot site; display delay of 10 seconds or less
Multi-user collaboration	Stated	Up to 12 concurrent users, on-site and remote mixed
Devices	Headsets plus tablet, phone and web	Specific headset models to be defined
Hybrid deployment, edge sync	Stated	On-site edge node; local operation offline for at least one 8-hour shift
Other features	Not stated	Everything marked Unverified in Section 5

2.3 Scope boundaries

The Product **is** a shared visual and collaboration layer, decision support, and a viewer of live and historical site state. It **is not** a replacement for planning suites, fleet management or SCADA, a certified safety system, or a source of legal, engineering or regulatory advice, unless and until proven or certified.

2.4 Commercial model (owner's choices)

- **Devices:** AR headsets plus tablet, phone and web access.
- **Deployment:** hybrid, with edge sync for poor connectivity.
- **Pricing:** a pilot fee, followed by an annual site licence.

---

3. Customers and Segments

No segment sizes are evidenced, and none should be quoted.

Segment	Core problem	Current alternative	Willingness to pay	Main barriers
Large open-pit producer	Disconnected planning, operations and safety data; slow reviews	2D plans, spreadsheets, meetings	Tier 1	IT/OT security, integration, change management
Underground operator	Complex 3D access, ventilation and rescue coordination	2D sections, paper plans	Tier 1–2	Harsh hardware conditions; connectivity (eased by edge sync)
OEM/dealer service	Costly expert callouts	Travel, phone and video	Tier 1–2	Needs equipment data access
Mining contractors	Client reporting, remote support	Site visits, photos, calls	Tier 2	Client owns the data
Mid-tier or junior miner	Limited staff and tools	Consultants, basic CAD	Tier 2–3	Budget, no data team
Insurers and lenders (technical advisors)	Remote physical-risk assessment	Site visits, reports	Tier 3	Data sharing from the miner
Training providers	Safe, realistic induction and drills	Classroom, site tours	Tier 3	Low budgets
Regulators and inspectorate	Inspection capacity, evidence	Site visits, paper	Tier 3	Government procurement, mandate

Willingness-to-pay tiers: 1 = likely budgeted, 2 = project-by-project, 3 = unclear or slow. All assessments are hypotheses to validate.

3.1 Prioritisation

Segments were scored 1–5 on pain, frequency, data fit, ability to pay, sales cycle (shorter scores higher) and strategic value. Scores are judgement-based.

Segment	Pain	Freq.	Data fit	Pay	Cycle	Strategic	Total
Large open-pit	4	5	4	5	2	5	**25**
Underground	5	4	3	4	2	4	**22**
OEM/dealer service	3	4	3	3	4	3	**20**
Contractors	3	4	3	3	3	3	**19**
Mid-tier	3	3	2	2	3	3	**16**
Insurers/lenders	3	2	2	3	2	3	**15**
Training	2	3	3	2	3	2	**15**
Regulators	3	3	2	1	1	4	**14**

Open-pit has the most data and money but long enterprise cycles. Underground has the highest pain, and edge sync reduces its data-fit risk. OEM service has a measurable cash baseline and a faster cycle. Regulators score low on payment and speed but matter strategically.

---

4. Buying Committee and Procurement

All entries are inferred or assumed and must be verified in discovery.

Segment	Economic buyer	Champion	Daily users	Security gate	Budget line
Large open-pit	GM or Head of Technical Services	Planning or innovation lead	Planners, geotech, HSE, maintenance	IT/OT security	Digital CAPEX or technical services OPEX
Underground	Mine manager	Ventilation or safety lead	As above, plus mine control	OT security	Safety or technical OPEX

Segment	Economic buyer	Champion	Daily users	Security gate	Budget line
OEM/dealer	Service manager	Field service lead	Service engineers	Customer IT	Service tooling
Contractors	Contracts or operations director	Site manager	Supervisors	Client IT	Project cost
Mid-tier	Owner or GM	Technical manager	Small technical team	Usually none	Project OPEX

Typical path: champion demo, site trial, technical and security review, paid pilot agreement, commercial approval, annual site licence, rollout. Group-level and government approvals may add months.

Common objections and honest responses

Objection	Response
"Our data won't integrate."	Scope one or two sources the site already exports. Promise no source until it is tested.
"Headsets won't survive the site."	Most users work on tablet, phone or web. Trial headsets in a bounded test.
"We have poor connectivity."	The edge node keeps sessions running locally and syncs later. Prove it in the pilot and measure offline duration.
"Where does our data live?"	Hybrid: data can stay on the edge node on site, with cloud sync by policy. Specifics require a security whitepaper.
"Who is liable if an alert is wrong?"	Alerts are decision support, showing latency and confidence. They do not replace certified systems.
"What does it cost versus save?"	The paid pilot measures the baseline. No return-on-investment claim is made before then.

5. Features and Maturity

Features marked "stated" were described by the product owner; no specification or demo has been seen, so none is confirmed as deployed.

Feature	What it does	Status	Known limitations
AR digital twin	Shows a 3D mine model in AR	Stated	Accuracy and model-update process unknown
Multi-device access	Same session across headset, tablet, phone, web	Stated	Supported models and feature parity unknown
Real-time streaming	Updates the twin with live data	Stated	Sources, latency and volume unknown
Multi-user collaboration	Shared sessions	Stated	Concurrency (assumed 12) unverified
Hybrid edge sync	Runs locally, syncs to cloud	Stated	Offline duration, conflict handling, hardware spec unknown
Geofence and proximity alerts	Warns on zone breaches	Unverified	Needs positioning hardware and safety validation
Slope and ground monitoring overlay	Shows movement against thresholds	Unverified	Not a substitute for geotechnical engineers
Ore model, grade overlay, plan vs actual	Puts model and blast data in context	Unverified	Import fidelity
Drone and LiDAR updates	Refreshes surfaces	Unverified	Processing time
4D playback and what-if	Replays the past, tests options	Unverified	Storage and fidelity
Fleet and SCADA integration	Links equipment status	Unverified	Vendor access
Work-order overlay, remote expert assist	Annotates equipment, enables remote help	Unverified	Bandwidth
AR training and emergency simulation	Practice scenarios	Unverified	Content effort

Feature	What it does	Status	Known limitations
Compliance records, audit trail	Permits, inspections, access logs	Unverified	Regulatory acceptance unknown
CAD, GIS and planning import	Brings existing models in	Unverified	Format coverage
Roles and data-residency options	Access control, data location	Unverified	Architecture unknown

6. Personas and Use Cases

No voice-of-customer data exists. All persona content is inferred or assumed.

6.1 Mine planning engineer

- **Job to be done:** align the plan with what actually happened and show it clearly.
- **Workflow:** weekly or shift review; load model, compare plan and actual, discuss with operations; produce a revised sequence and action list; decide what to mine or move next.
- **Pain points:** scattered data, slow shared understanding, manual reports built from screenshots and slides.
- **Features used:** AR twin, multi-user, streaming; plan vs actual and fleet integration (unverified).
- **Benefits:** time saved (soft) and shared understanding (qualitative).
- **Must not claim:** replacement of the planning suite; improved recovery or grade.
- **Pilot metrics:** review duration, preparation hours, actions closed on time.

6.2 HSE manager

- **Job to be done:** reduce exposure and evidence compliance.
- **Workflow:** after an incident, audit or new work area, review the area in 3D, brief crews, update exclusion zones, record the outcome.
- **Pain points:** hard-to-visualise hazards, manual induction, hand-compiled evidence.
- **Features used:** AR twin, multi-user; geofence alerts, training and audit trail (unverified).

- **Benefits:** induction time (soft) and clearer hazard communication (qualitative). Risk reduction is contingent and never added to savings.
- **Must not claim:** that it prevents incidents, is a certified safety system, or satisfies regulatory inspection.
- **Pilot metrics:** induction hours, briefing completion, alert latency (if built), clarity score.

6.3 Maintenance planner and reliability engineer

- **Job to be done:** shorten downtime and avoid repeat failures.
- **Workflow:** on breakdown or shutdown, locate the asset, connect an expert, annotate, record the repair, decide whether to escalate.
- **Pain points:** waiting for experts, poor remote communication.
- **Features used:** multi-user, AR; remote assist and CMMS overlay (unverified).
- **Benefits:** avoided callouts (hard, if the cost is external) and shorter downtime (needs measurement).
- **Must not claim:** predictive-maintenance accuracy or integration with any named maintenance or fleet system until verified.
- **Pilot metrics:** callouts avoided, time to repair, expert wait time.

6.4 Geotechnical engineer

- **Job to be done:** turn monitoring data into access decisions that operations will act on.
- **Workflow:** on a movement alert or review, examine monitoring data and models in context with operations, then issue an advisory or exclusion recommendation.
- **Pain points:** monitoring data sits in separate tools; movement is hard to explain to operations in 2D.
- **Benefits:** faster shared context (qualitative).
- **Must not claim:** early warning, stability prediction, or replacement of engineering judgement.
- **Pilot metric:** time from alert to operations briefing.

6.5 Other personas

Persona	Role of the Product	Must not claim	Pilot metric
Mine manager or GM	Overview and decision forum	Production uplift	Review attendance, decision latency
IT/OT security	Gatekeeper	"Secure" without audit evidence	Security review pass
Inspector or regulator	Remote review, evidence viewing	Regulatory acceptance	Inspection preparation time
Insurer or lender advisor	Remote site view	Risk-rating accuracy	Visit-days avoided
OEM service engineer	Remote assist with site context	Fault-diagnosis accuracy	Callouts avoided

7. Business Value Model and Pricing

All figures are **illustrative assumptions** to be replaced with pilot data. Hard savings (cash), soft savings (redeployed time) and contingent avoidance (probabilistic) are shown separately and are never summed.

7.1 Formulas

Use case	Type	Formula
Planning review	Soft	reviews per year × participants × hours saved × loaded hourly cost × adoption
Remote assist (maintenance or OEM)	Hard, if external	callouts per year × avoided fraction × cost per callout × adoption
HSE induction	Soft	inductions per year × hours saved × loaded hourly cost × adoption
HSE incident risk	Contingent	(baseline probability – assisted probability) × cost per event
Mine manager	None	No formula; there is no basis for a production-uplift claim

7.2 Assumed inputs

Variable	Low	Base	High
Reviews per year	50	100	200
Participants per review	5	8	12
Hours saved per person per review	0.25	0.5	1.0
Loaded hourly cost (USD)	25	40	60
Adoption	30%	60%	80%
Callouts per year	15	30	60
Avoided fraction	20%	40%	60%
Cost per callout (USD)	800	1,500	3,000
Inductions per year	300	600	1,200
Hours saved per induction	0.5	1.0	1.5

7.3 Results per site per year (USD, rounded)

Use case	Type	Low	Base	High
Planning review	Soft	470	9,600	115,200
Remote assist	Hard, if external	720	10,800	86,400
HSE induction	Soft	1,125	14,400	86,400

Base-case workings: planning 100 × 8 × 0.5 × 40 × 0.6 = 9,600; remote assist 30 × 0.4 × 1,500 × 0.6 = 10,800; induction 600 × 1.0 × 40 × 0.6 = 14,400.

Sensitivity. Halving both hours saved and adoption in the planning example cuts value to 2,400, a quarter of base. Halving the avoided fraction in remote assist cuts value to 5,400.

Contingent example (not a saving). (2.0% – 1.5%) × USD 500,000 = USD 2,500 expected value per year, using placeholder probabilities.

7.4 Pricing model and break-even

The commercial model is a pilot fee followed by an annual site licence. Price points are assumptions to test.

Item	Low	Base	High
Pilot fee (8–12 weeks, one site, up to 15 users)	10,000	15,000	25,000
Annual site licence	24,000	36,000	48,000
Edge node hardware, one-off, if client-bought	8,000	12,000	15,000

Base-case coverage of the USD 36,000 licence, one use case at a time

Use case	Base value	Share of licence
Remote assist (hard)	10,800	30%
Planning review (soft)	9,600	27%
HSE induction (soft)	14,400	40%

What this means

1. No single base-case use case justifies the base licence, and soft savings are weaker than cash.
2. Hard value covers the licence only at high callout volumes. In the high case, remote assist yields 86,400, above even the high licence of 48,000. This points to OEM, dealer and high-callout operations as the first buyers.
3. For producers, the pitch needs several use cases per site, a licence nearer USD 24,000, or pricing that scales with sites and users.
4. Crediting the pilot fee against the first licence year would reduce buyer risk.

What must be true: data feeds integrate and stay fresh; the model stays accurate; users adopt the Product beyond the demo; hardware works on site; edge sync keeps sessions usable offline; and security approval is granted.

---

8. Competitive and Substitute Landscape

This section is category-level only. No web research was done and no vendor facts are asserted. All comparisons are hypotheses.

Alternative	Strength	Likely weakness against the Product	Switching cost
Status quo (2D plans, meetings, spreadsheets)	Free and familiar	Low shared context	Low
Mine planning and geology suites	Deep technical function	Not built for shared AR	Medium
Fleet-management and SCADA dashboards	Operational data	Limited spatial collaboration	Medium
General digital-twin platforms	Broad scope	Mining depth, local support, offline operation	High
Generic AR/VR or video-assist tools	Remote help	No live mine model	Low
Consultants and integrators	Custom work	Cost and dependence	Medium

To complete: a researched comparison covering Ghana and West Africa coverage, auditability, price signals and local support, with sources and access dates.

---

9. Positioning and Messaging

Segment	Positioning	Proof points	Do not say
Large open-pit	"See the same live mine model, together, wherever you are."	AR twin, streaming, multi-user, multi-device (stated)	"Improves recovery," "reduces incidents," "replaces your planning suite"
Underground	"A shared 3D picture for ventilation, access and safety coordination, even when the connection drops."	Twin, multi-user, edge sync (stated); sensor overlays (unverified)	"Gas or ventilation monitoring," "rescue-grade"
OEM/dealer	"Fix it with the expert looking at the same	Multi-user, multi-device (stated);	"Predictive maintenance"

Segment	Positioning	Proof points	Do not say
	machine and site."	remote assist (unverified)	
Contractors	"One shared view for you and your client."	Multi-user, web access (stated)	"Client-approved reporting"

Sales material that can be produced now: a pilot proposal with the paid-pilot terms, a 15-minute demo script on one site model, a one-page integration questionnaire, an edge-node requirements sheet, and a pilot KPI sheet. Outbound material must contain no quantified savings until pilots produce them.

---

10. Local Context: Ghana

The points below come from general knowledge, sit outside the owner's inputs, and **require legal verification** before use.

- **Licensing:** the Minerals and Mining Act, 2006 (Act 703) governs mineral rights, and the Minerals Commission administers licences and policy.
- **Inspection and safety:** mine health-and-safety rules and the Inspectorate function sit apart from licensing. Check current regulations before claiming any compliance use.
- **Environmental permitting:** confirm the role of the Environmental Protection Agency.
- **Data protection:** the Data Protection Act, 2012 (Act 843) may apply to worker, position and identity data. Hybrid deployment helps in residency discussions.
- **Cybersecurity:** check current obligations for critical operations under Ghana's cybersecurity legislation.
- **Illegal small-scale mining (galamsey):** it affects concession security and may create demand for boundary monitoring, but this is not a stated Product capability.
- **Community and land issues:** these affect operations and site access.
- **Infrastructure:** unreliable power and connectivity, and hardware import costs, constrain AR. The edge node and multi-device access are the mitigations, and the edge specification should include a power-backup requirement.
- **Regional expansion:** consider after Ghana pilots; nothing has been gathered yet.

---

11. Risks, Dependencies and Open Questions

Risk	Likelihood	Impact	Mitigation	Suggested owner
Feature maturity overstated	Medium	High	Specification audit; feature-gated sales material	Product
Integration effort exceeds pilot scope	High	High	Scope 1–2 sources; connector roadmap	Engineering
Alert latency or false alarms	Medium	High	Show latency and confidence; supplement, never replace, certified systems	Product, Safety
Liability for wrong outputs	Medium	High	Contract terms, disclaimers, legal review	Legal
Headset unsuitable on site	Medium	Medium	Multi-device access; hardware trial	Engineering
Edge sync fails after long offline periods	Medium	High	Test 8 hours offline; define conflict rules	Engineering
Model accuracy or currency lapses	Medium	High	Show data timestamps; define update process	Product
Security or data-residency blockers	Medium	High	Security whitepaper; edge-resident data option	IT, Security
Licence not covered by measured value	High	High	Multi-use-case pitch; OEM-first; tune price after pilot	Commercial
Government procurement	High	Medium	Deprioritise regulators	Sales

Risk	Likelihood	Impact	Mitigation	Suggested owner
delays				
Low sustained adoption after pilot	Medium	High	Embed in an existing meeting; track usage	Customer success

12. Validation and Pilot Plan

12.1 Hypotheses

All hypotheses start as unvalidated assumptions. Thresholds are placeholders.

#	Hypothesis	Test	Success threshold
H1	Shared AR review is shorter or clearer than current meetings	Timed pilot	20% shorter, or clarity score of 4/5 or better
H2	Weekly review is the strongest entry use case	Interviews	6 of 10 name it a top-3 pain
H3	Remote assist avoids a quarter of expert callouts	Pilot log	25% or more
H4	Producers will pay for a pilot	Offers	2 or more paid pilots
H5	1–2 data sources can be integrated within 4 weeks	Pilot	Delivered on time
H6	Tablet, phone and web suffice for most users	Usage log	70% or more of sessions non-headset
H7	Underground is viable with edge sync	Site test	Stable session for 8 hours offline, clean sync after
H8	HSE values visual hazard briefing	Trial	20% less induction time
H9	IT/OT approves under a defined security posture	Security review	Approval within 8 weeks

#	Hypothesis	Test	Success threshold
H10	Use is sustained after the pilot	Usage tracking	60% of pilot users active weekly at week 8
H11	OEM and dealer teams buy faster than producers	Pipeline	Cycle at least 30% shorter
H12	Buyers accept a pilot fee then site licence	Pricing test	50% or more of pilots convert
H13	Measured value covers the low licence (USD 24,000)	Pilot data	Hard and soft value reported separately; hard covers at least 50%

12.2 Pilot template

Element	Content
Commercial	Paid pilot, base USD 15,000, credited against the year-one licence on conversion
Scope	One site, one use case (review or remote assist), 1–2 data sources, up to 15 users, one edge node
Baseline	2–4 weeks before launch: meeting durations, preparation hours, callouts, induction hours
Instrumentation	Session logs, time stamps, short survey, callout log, offline and sync log
Duration	8–12 weeks
Exit criteria	Agreed thresholds met, usage sustained, security approved
Outputs	Placeholders replaced with measured values; hypotheses re-tiered; licence price set

12.3 KPI dictionary

KPI	Definition	Source
Review duration	Start-to-end of a scheduled review	Calendar and session log

KPI	Definition	Source
Preparation hours	Hours spent preparing review materials	Time survey
Callouts avoided	Planned callouts resolved remotely	Service log
Time to repair	Fault reported to resolved	Maintenance system
Induction hours	Hours per new worker	Training records
Active users	Users with at least one session per week	Platform logs
Non-headset share	Non-headset sessions ÷ all sessions	Platform logs
Data latency	Event-to-display delay	Platform logs
Offline resilience	Hours a session stays usable offline; sync success rate	Edge logs
User clarity score	1–5 survey rating	Survey

12.4 Discovery interview guide (30–45 minutes, eight questions per segment)

Large open-pit: (1) Walk me through your last weekly review. (2) What data do you open, and from which systems? (3) Where does the time go? (4) How do you share the plan with operations today? (5) Who must approve new software? (6) What blocked your last technology rollout? (7) What do you spend on this today? (8) What would make you stop using a tool after a month?

Underground: (1) How do you coordinate access and ventilation changes? (2) What sensor data do you get, and how often? (3) What connectivity do you have underground? (4) What hardware is allowed? (5) How do you brief crews on hazards? (6) What does an emergency drill involve? (7) Who signs off safety tools? (8) What has failed before?

OEM and dealer service: (1) How many expert callouts did you make last year? (2) What does a callout cost in travel and time? (3) Which fault types need an expert? (4) What remote tools do you use now? (5) What do you need to see to help remotely? (6) Who approves tools? (7) Whose data is it? (8) What would a trial need to show?

13. Roadmap Implications

Rank	Item	Rationale	Segments
1	Data connectors (fleet management, survey and model import)	Gates adoption and value everywhere	All
2	Edge-sync hardening and an edge-node specification	Underground and Ghana-site viability depend on it	Underground, remote sites
3	Remote expert assist with session records	Clear cash baseline, fastest sale, best hard-value case	OEM, maintenance
4	Plan vs actual overlay	Core to the weekly review	Open-pit
5	Audit trail, roles, data-residency options	Unblocks security review; fits the hybrid story	All enterprise
6	Usage analytics for pilots	Needed to measure value and price the licence	All
7	Alerts (geofence, ground)	High value but high liability; validate first	HSE, geotech
8	4D playback and what-if	Useful, lower urgency	Planning
9	AR training	Smaller budgets	Training, HSE

14. Evidence Base, Assumptions and Open Items

14.1 Evidence tiers

- **Documented:** confirmed in product documentation. None exists yet.
- **Planned:** documented as partial or planned. None exists yet.
- **Inferred:** reasoning stated in this document.
- **Stated by owner:** described in chat, with no specification or demo seen. This is not documentation-grade evidence.

- **Assumption:** a hypothesis to validate. Every assumed number is a placeholder.

14.2 Evidence summary

Claim	Source	Tier
AR digital twin, real-time streaming, real-time multi-user collaboration	Owner chat	Stated by owner
Headsets plus tablet, phone and web; hybrid deployment with edge sync; pilot fee then annual site licence	Owner chat (option selections)	Stated by owner
Weekly review is a strong fit; remote assist has a cash baseline	Reasoning from stated features	Inferred
12 concurrent users, 10-second latency, 0.5 m alignment, 8-hour offline operation	Planning assumptions	Assumption
All price points and value inputs	Planning assumptions	Assumption
Ghana legal items	General knowledge, outside inputs	Verify

14.3 Assumptions to confirm

1. The Product is sold to mining operators first, with OEM and dealer service as a fast-sale second channel.
2. The owner's three stated features and three option choices stand until a specification or demo exists.
3. Headset models, edge hardware and per-seat limits are unspecified and need a device matrix.
4. Data sources and protocols are unspecified; pilots start with one or two.
5. Price points are placeholders, to be set after pilots.
6. Ghana is the launch market.

14.4 Known gaps

There is no voice-of-customer data, no external research, no product specification, and no researched competitor comparison. The original brief described a different geospatial-intelligence product (MINERVA, MINEASK, GeoMine); that scope was set aside at the owner's instruction, and its consultant, lender, regulator and ESG personas were replaced with operations personas.

14.5 Glossary

AR: augmented reality. **ASM:** artisanal and small-scale mining. **CMMS:** computerised maintenance management system. **Concession:** a licensed mining area. **Digital twin:** a live digital model of a physical asset. **Edge node:** a local server at the mine that keeps the system running and syncs to the cloud. **EITI:** Extractive Industries Transparency Initiative. **FMS:** fleet management system. **Galamsey:** Ghanaian term for illegal small-scale mining. **OT:** operational technology. **SCADA:** supervisory control and data acquisition.